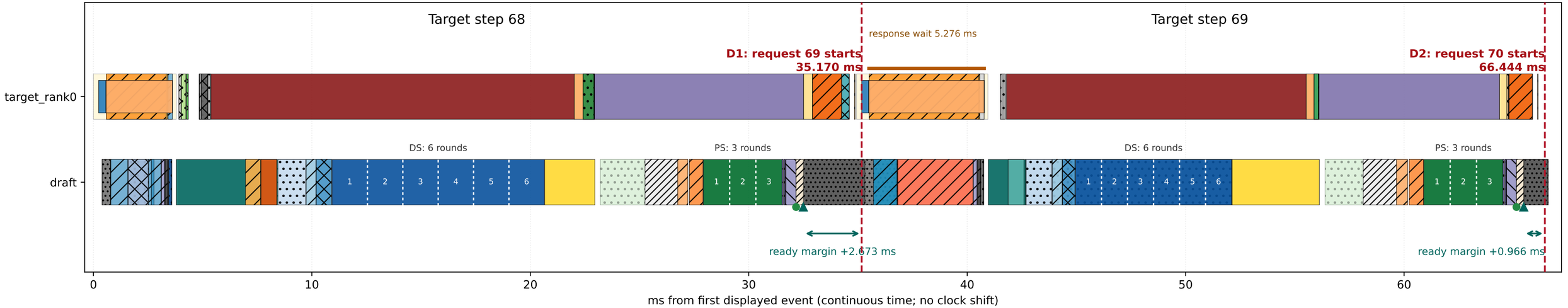
LayerSkip Llama2-13B + AMD-Llama-135M | Selected setting | P1 hit | UID 84_t0 | steps 68-69

Historical v1.0 59f46511 | exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
68	32.167	32.497	35.170	+3.003	+2.673	0.314	5.276
69	65.147	65.478	66.444	+1.297	+0.966	0.321	2.856

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

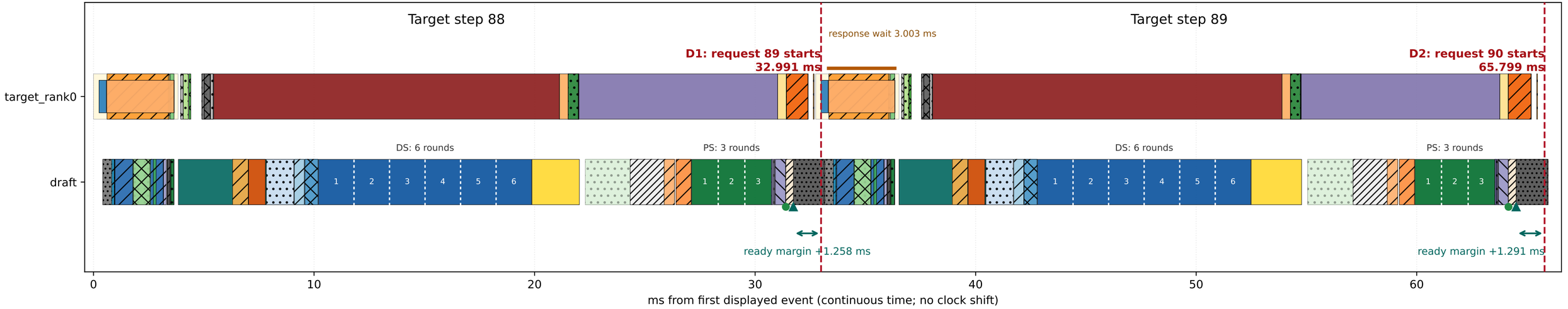
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip Llama2-13B + AMD-Llama-135M | Selected setting | P2 hit | UID 244_t0 | steps 88-89

Historical v1.0 59f46511 | exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
88	31.394	31.733	32.991	+1.597	+1.258	0.322	3.003
89	64.164	64.508	65.799	+1.635	+1.291	0.346	2.953

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

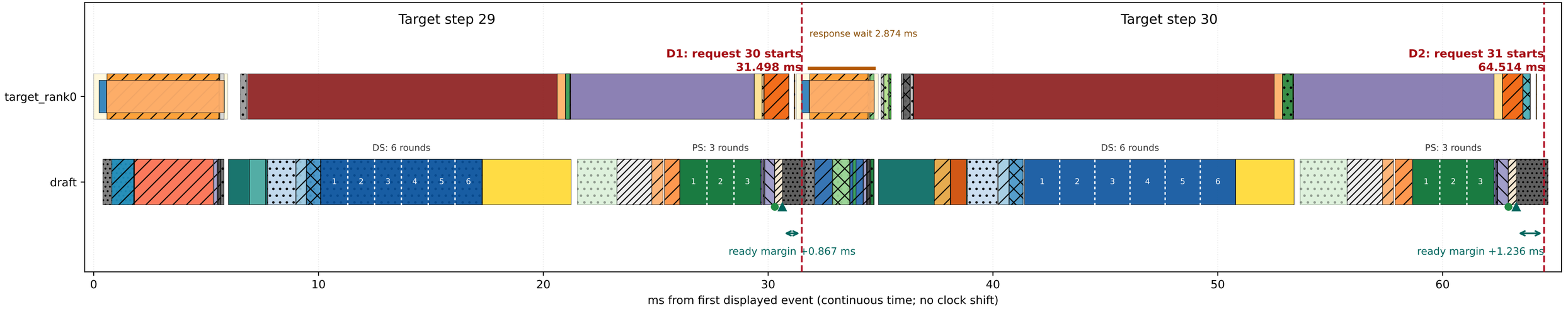
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

LayerSkip Llama2-13B + AMD-Llama-135M | Selected setting | Miss | UID 128_t0 | steps 29-30

Historical v1.0 59f46511 | exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.



Prepared in step	Cache ready (ms)	Receive-ready (ms)	Next send (ms)	Cache margin (ms)	Ready margin (ms)	Next send span (ms)	Next response wait (ms)
29	30.298	30.631	31.498	+1.200	+0.867	0.322	2.874
30	62.933	63.277	64.514	+1.581	+1.236	0.347	3.728

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.